and completeness, but Bell's impossibility proof showed that the choice is actually between locality and *correctness.* Starting from the assumption that nature is local, Bell derived an inequality, a mathematical condition that any local theory of nature has to meet. Then Bell shrewdly altered Bohm's version of the EPR thought experiment to create a situation where the predictions of quantum physics violate that inequality.

Bell's stroke of brilliance was to consider imperfection, rather than perfection. After all, the perfect correlations in the EPR-Bohm setup are easily compatible with locality—the photons could be sharing hidden instruction sets at their common origin. But if you rotate the axis of one of the polarizers, quantum physics predicts that pairs of entangled photons arriving at the polarizers will no longer behave in exactly the same way every time. And Bell showed that the imperfect correlations predicted by quantum physics were too strong for any local theory of nature to be able to account for them. So either the predictions of quantum physics are wrong and nature can be local, or quantum physics is right and "spooky action at a distance" is real. Bell had discovered a remarkably profound and counterintuitive truth about the world.

Bell had also shown that there was an experimental test that could decide between the two options. All someone had to do was actually construct and perform Bell's modified version of the EPR thought experiment, or another experiment along those lines involving entangled particles. If the results showed that Bell's inequality was violated, quantum physics was safe but nature was nonlocal; if his inequality held, then quantum physics was wrong but nature could be local. Bell's impossibility proof had taken the question of nonlocality out of the realm of debate and turned it into an experimental challenge. This proof, now known as Bell's theorem, has rightly been called "the most profound discovery of science."

Bell's result is both unexpected and problematic. Locality is a basic assumption of physics, and indeed all of science. Without locality, it would be difficult to perform any controlled experiments at all—no matter how well you control the surroundings of your experiment, there would always